VIDEO GAME CREATION

Lesson 10 - Core Mechanics IV: Play It Again

Semester 1, Week 10 - Game-Over + Restart Flow

- **Objectives:** Refactor existing one-time setup code into a reusable reset routine, callable from two different entry points; introduce a game-state flag that gates whether gameplay logic still applies; implement a full, clean restart triggered by a dedicated key press.
- **IGDA Tie-In:** Programming - state machines (introductory), code reuse/refactoring, input handling (intermediate)
- **Featured Engine (lecture demo):** Godot Engine
- **Materials:** Demo laptop with all 8 engines verified (see Teacher's Checklist); Student Handouts, one per engine; students' own Lesson 9 projects (already saved) - no new assets needed this week

Agenda

0:00-0:05	Welcome back, Lesson 9 recap
0:05-0:20	The Last Missing Piece: What 'Over' Should Actually Mean
0:20-0:40	One Setup, Two Doors In (live demo in Godot)
0:40-0:55	The Freeze: A Flag That Gates Everything
0:55-1:00	Introduce lab exercise, hand out engine-specific handouts
1:00-1:50	LAB: Build game-over freeze and restart flow
1:50-2:00	Exit check / quick share-out - full playthrough demos

Levels for this lesson's exercise

- Level 1 (Minimal): when the round ends, the game freezes; pressing R resets everything and starts a new round.
- Level 2 (Complete): a restart is a genuinely clean slate - no leftover enemies, no stale score/lives, works correctly any number of times in a row.
- Level 3 (Refined): add a transition effect, and/or tighten the single-source-of-truth structure of the reset routine.

See: Lesson 10 Teacher's Script, Lesson 10 Prep Checklist, and the per-engine Student Handouts. Every engine involves a genuine refactor this week (existing setup code restructured into a reusable form), not just new additions.